Nikoo Vali

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Education

University of British Columbia, *Bachelor of Applied Science, Computer Engineering* ***Expected: December 2027***
Achievements: Outstanding International Student Award , Undergrad Research Award (2025)
Courses: *Software Construction I , II , Data Structure and Algorithms, Machine Learning, Operating Systems*

Experience

Seaspan **May 2026 – Present**
Data Engineer Intern *Vancouver, BC*

- Built **Spark** ETL pipelines in **Databricks** using **SQL** and **Python**, transforming raw operational data into analytics-ready datasets
- Automated data ingestion, validation and orchestration in **Azure Data Factory**, improving reliability across data workflows
- Developed API-driven ingestion pipelines to extract and load enterprise data into **Azure Data Lake** for downstream analytics
- Normalized and Flattened nested JSON API payloads into structural relational datasets for analytics and enterprise reporting
- Designed analytics-ready data models and transformations for **Power BI**, enabling new operational metrics for business teams

National Research Council Canada **Sep 2025 – Apr 2026**
Software Engineer - Research Assistant *Vancouver, BC*

- Developed a Python validation pipeline with **Pandas/regex**, improving record-level validation and reducing processing latency
- Engineered a semantic data-matching system in Python with NumPy automating cross-database record alignment and reconciliation
- Implemented a content-hash cache in Python using **hashlib (SHA-2)** and pickle serialization, cutting repeated runtime by **90%**
- Designed an incremental processing pipeline to recompute only new or modified records, eliminating full-dataset reprocessing

UBC Centre of Technology **May 2025 - Aug 2025**
Software Engineer Intern *Vancouver, BC*

- Built PostgreSQL validation tests to verify transactional persistence, retrieval and data consistency across application workflows
- Automated API testing using **PyTest** to verify REST endpoints, HTTP responses, and backend service functionality
- Created regression test suites to detect breaking changes in authentication, routing, and error handling across **FastAPI** services

Digital Solutions **Jan 2025 - May 2025**
Software Engineer Intern *Vancouver, BC*

- Integrated cloud-backed storage pipelines with **AWS S3** to support reliable object storage, retrieval, and application data workflows
- Containerized backend microservices with **Docker** and PostgreSQL deploying automated API workflows to support **1,500+ users**

UBC Open Robotics **May 2024 - Sep 2025**
Software Engineer – NLP *Vancouver, BC*

- Engineered a modular **NLU pipeline** for intent parsing and fallback recovery, improving command reliability by **30%**
- Designed a vision-based perception pipeline using **YOLO** for real-time object detection, classification, and spatial localization
- Processed unstructured voice commands into structured intents and entities for reliable downstream action routing

Technical Skills

Languages: Python, SQL, C++, C, TypeScript, C#, Java, Kotlin, JavaScript, Golang, Ruby, Bash, HTML/CSS, YAML
Technologies: Databricks, Apache Spark, FastAPI, Flask, PostgreSQL, Azure Data Factory, PyTorch, TensorFlow, Pandas
Developer Tools: Git, Docker, CI/CD, PyTest, Playwright, Selenium, Jenkins, Unix/Linux, Postman, Jupyter Notebook, Jira

Projects

CryptoPulse: Market Sentiment & Price Prediction Platform (Link) | *Python* *Personal Project* | July 2025

- Built **Python/pandas** pipelines to ingest and clean crypto price and social-media data for downstream NLP and forecasting
- Applied **spaCy and Hugging Face Transformers** for sentiment classification and generation of time-aligned text features
- Engineered market and sentiment features for predictive modeling from historical price data and NLP-derived signals
- Implemented data-quality checks for missing values, outliers, and timestamp mismatches to improve downstream model reliability

MindLink: Therapy Training Platform (Link) | *Python, Flask, React* July 2025

- Built an AI-powered therapy training platform using Python (Flask) **React** and **OpenAI** APIs to simulate patient interactions
- Designed relational data models for session transcripts, user interactions, and evaluation metrics using SQLAlchemy
- Built analytics workflows to summarize trainee performance and generate actionable feedback from conversation data

Crushing Coding Interviews(Link) | *TypeScript, Express.js, MongoDB, OpenAI, Kotlin* October 2025

- Built a full-stack Android application in **Kotlin** for AI-driven behavioral and technical interview preparation
- Designed a modular **REST API (Express.js)** for users, jobs, questions and discussions with Google OAuth authentication
- Implemented scalable data models using **MongoDB**, supporting user profiles, sessions, and progress tracking